

zomato

(An Exploratory Data Analysis
of Indian Restaurants)

by - Mayank



Introduction

The Zomato logo is displayed in white, lowercase, bold letters on a red rounded square background.

Overview of the project

This project focuses on analyzing Indian restaurant data sourced from Zomato to understand the factors that influence restaurant success and customer satisfaction. The analysis is based on restaurant ratings, customer votes, price range, cuisines, and city-wise distribution.

The objective of this project is to extract meaningful insights from the data that can help restaurant owners and food platforms understand customer preferences and make better business decisions.

Objectives

- To analyze Indian restaurant data sourced from Zomato
- To study restaurant ratings and customer satisfaction
- To examine the relationship between ratings and customer votes
- To analyze the impact of price range on restaurant ratings
- To identify popular cuisines and city-wise restaurant trends
- To extract insights that support data-driven business decisions



EDA step by step



Step 1: Importing Required Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

EDA begins by importing Python libraries such as pandas and numpy for data manipulation, and matplotlib and seaborn for data visualization.

Purpose:

These libraries help in efficiently handling data and creating meaningful visualizations.



EDA step by step



Step 2: Loading the Dataset

```
df=pd.read_csv("/content/Indian-Restaurants.csv")
```

The dataset is loaded using pandas and the first few records are inspected.

Purpose:

To understand the type of data available and verify successful data loading.

Observation:

The dataset contains restaurant-related attributes such as city, rating, votes, price range, and cuisines.



EDA step by step



Step 3: Understanding Dataset Structure

```
df.head()
df.shape
df.info()
df.isnull().sum().sort_values(ascending=False)
```

Functions like `.info()`, `.shape()`, and `.isnull()` are used.

Purpose:

To understand the size of the dataset, data types, and missing values.

Observation:

The dataset contains a large number of records with both numerical and categorical variables. Missing values were mainly found in the cuisine column.



EDA step by step



Step 4: Data Cleaning

```
drop_cols = [  
    'res_id', 'city_id', 'country_id',  
    'url', 'address', 'zipcode', 'timings', 'highlights',  
    'photo_count', 'opentable_support',  
    'latitude', 'longitude'  
]  
df = df.drop(columns=drop_cols, errors='ignore')
```

Irrelevant columns such as IDs, URLs, coordinates, and address details were removed.

Purpose:

To focus the analysis only on business-relevant features.



EDA step by step



Step 5: Statistical Summary

```
df.describe(include='all')
```

Descriptive statistics such as mean, median, minimum, and maximum were analyzed.

Purpose:

To understand the distribution and central tendency of numerical variables like ratings and votes.



EDA step by step



7.CORRELATION ANALYSIS

```
df[['aggregate_rating', 'votes', 'price_range']].corr()
```

	aggregate_rating	votes	price_range
aggregate_rating	1.000000	0.276279	0.253581
votes	0.276279	1.000000	0.249013
price_range	0.253581	0.249013	1.000000

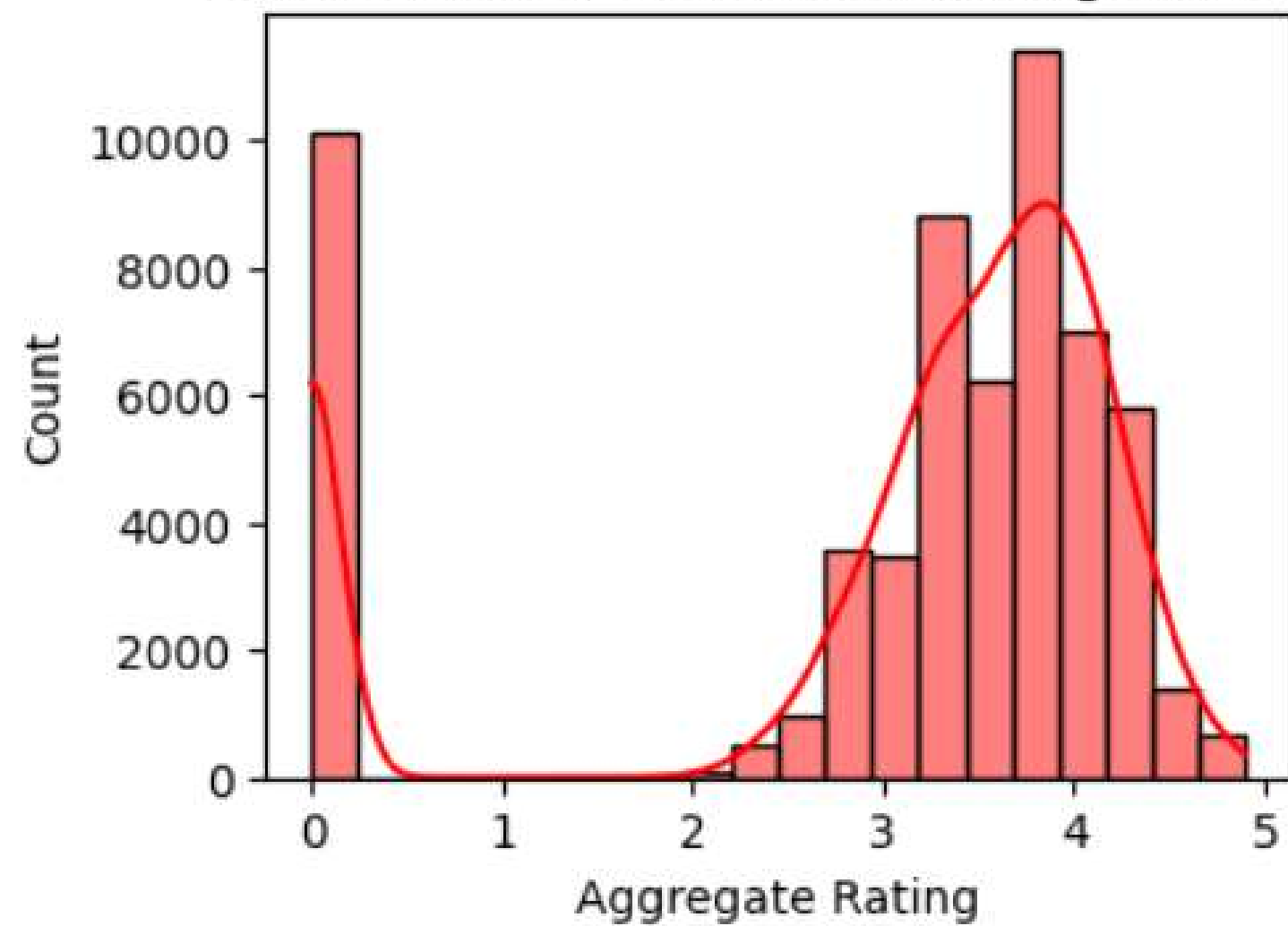
Although ratings, votes, and price range show a positive relationship, the low correlation values indicate that none of these factors alone strongly determines restaurant success. Instead, success is influenced by multiple factors working together.



visualization



Distribution of Restaurant Ratings in India



1. Distribution of Restaurant Ratings

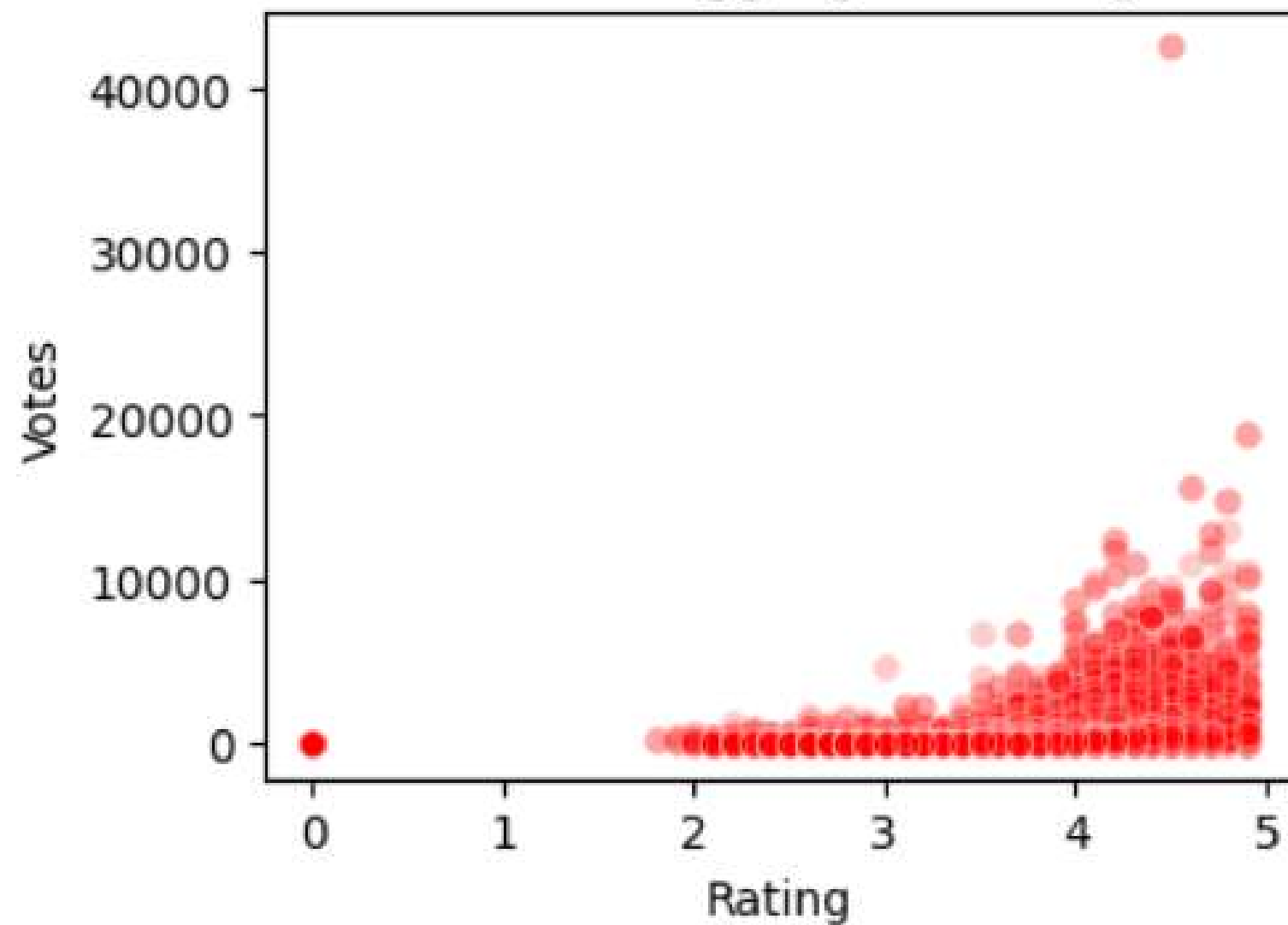
This graph shows how restaurant ratings are spread across the dataset. Most restaurants have ratings between 3.5 and 4.5, indicating generally good customer satisfaction. Very few restaurants fall in the low-rating range.



visualization



Votes vs Aggregate Rating



2. Votes vs Aggregate Rating

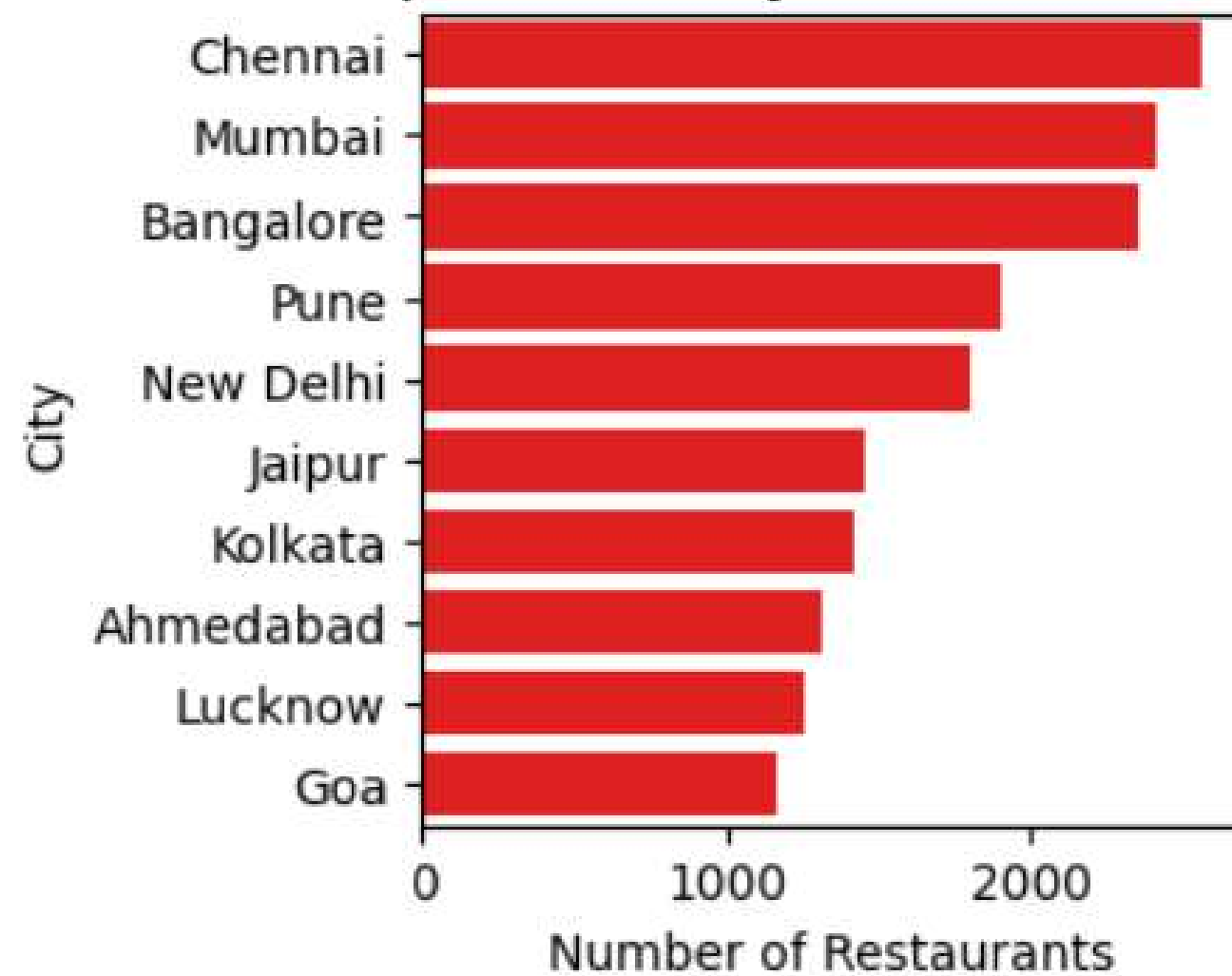
This scatter plot shows the relationship between ratings and customer votes. Restaurants with more votes are generally more trusted, even if their ratings are slightly lower. Ratings with very few votes may be less reliable.



visualization



Top 10 Cities by Restaurant Count



3. City-wise Restaurant Distribution

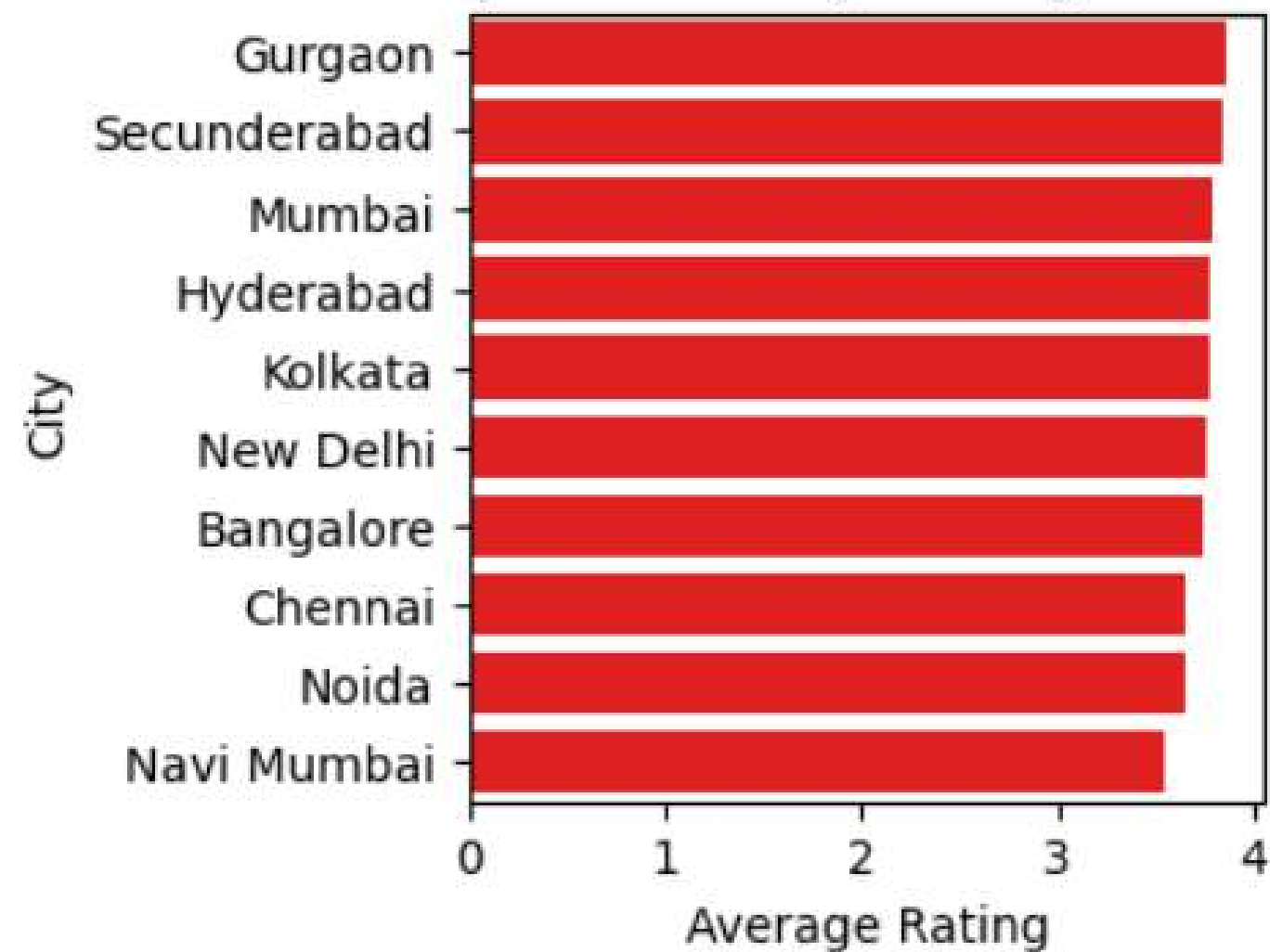
This graph shows cities with the highest average ratings. Some smaller cities perform better than metro cities, indicating higher service consistency and customer satisfaction.



visualization



Top 10 Cities by Average Rating

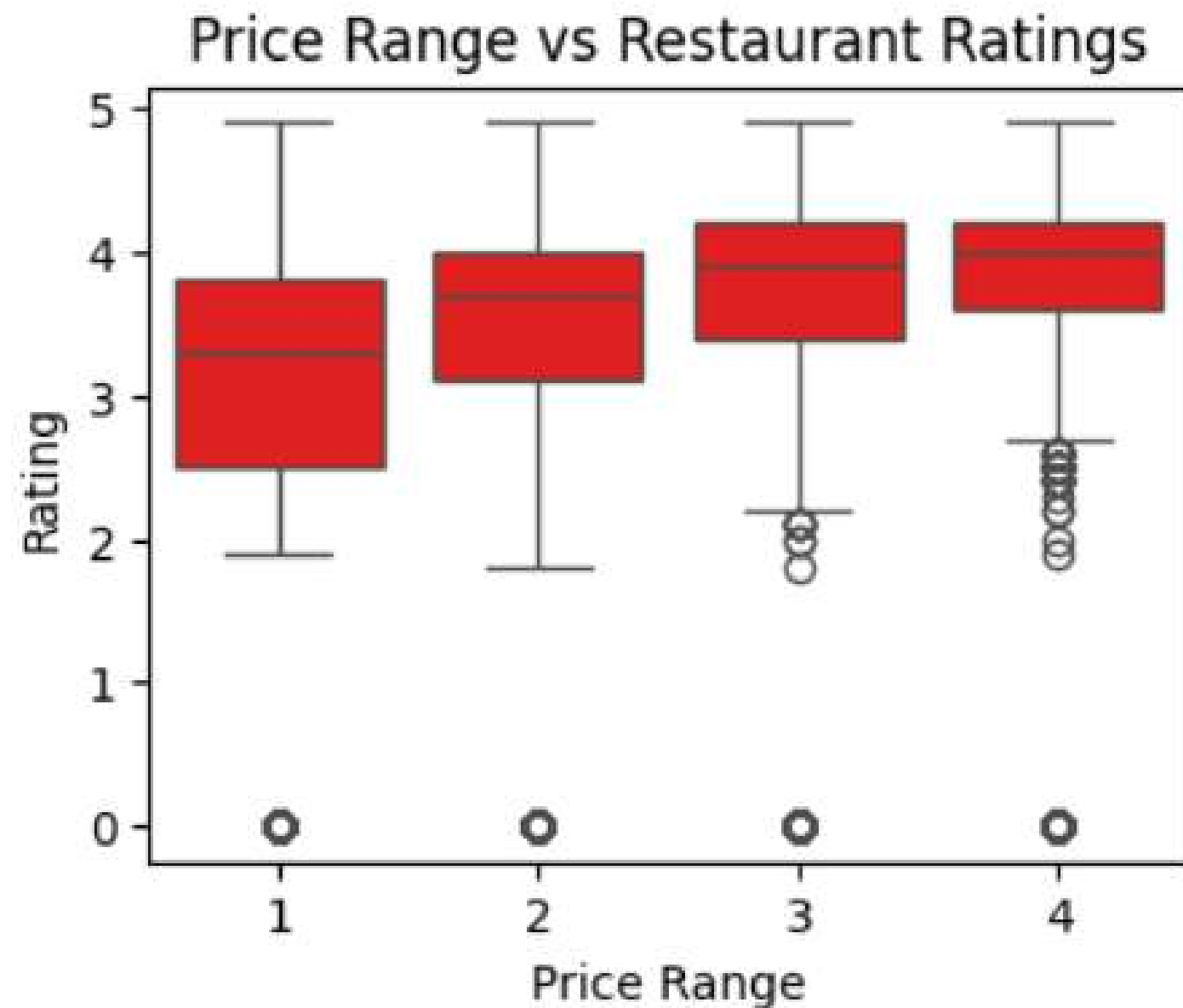


4. City-wise Average Restaurant Rating

Since this visualization includes only the top 10 cities by average rating, the ratings appear very similar and fall within a narrow range. This suggests that among high-performing cities, customer satisfaction levels are consistently high, with no single city having a strong advantage over the others.



visualization



5. Price Range vs Restaurant Ratings

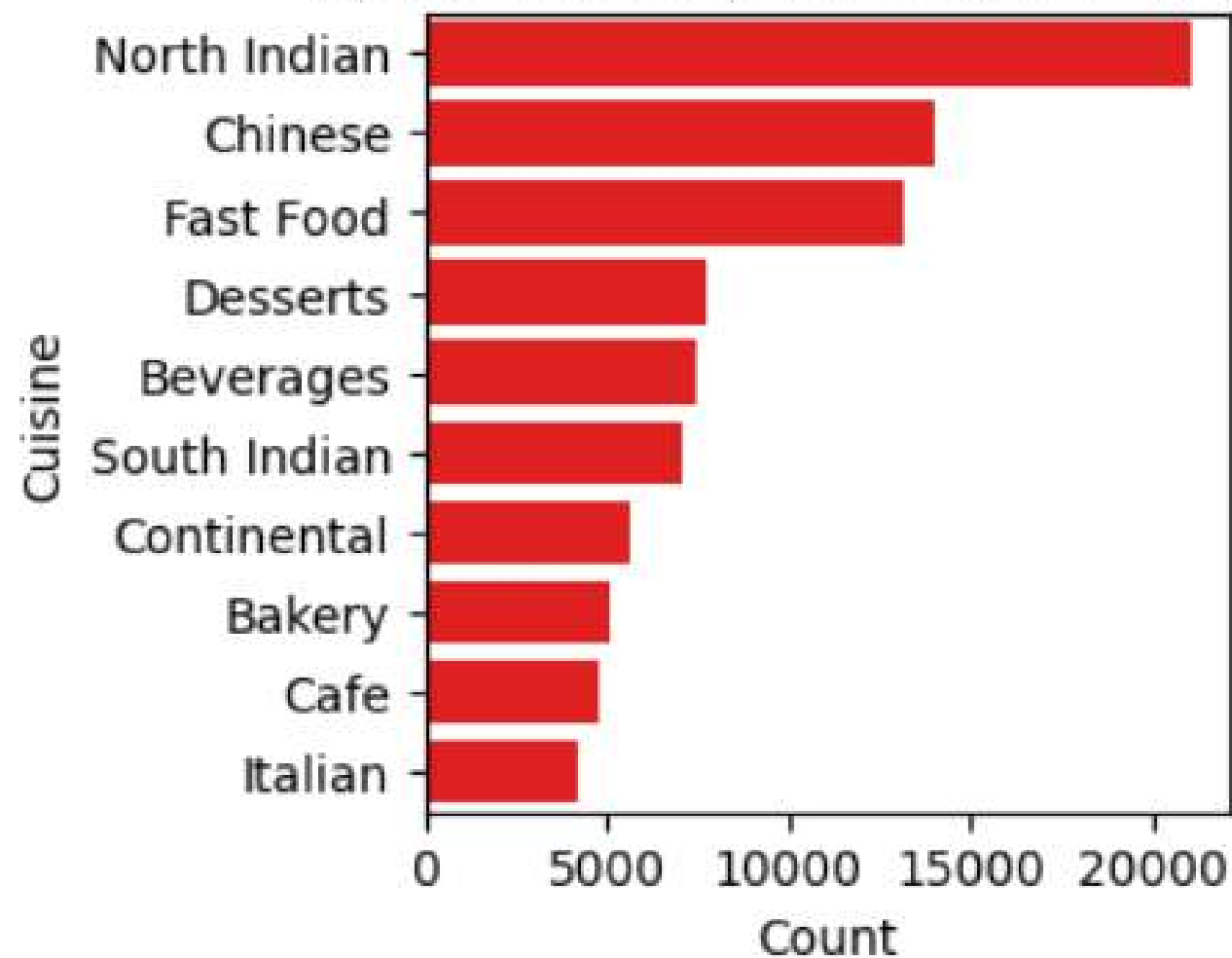
This box plot compares ratings across different price ranges. Higher-priced restaurants show only a slight improvement in ratings, suggesting that higher prices do not guarantee better customer satisfaction.



visualization



Top 10 Most Popular Cuisines in India



6. Cuisine Popularity Analysis

This bar chart highlights the most popular cuisines in India. North Indian and Chinese cuisines dominate in volume, while some niche cuisines tend to receive better ratings.



INSIGHT



- Most restaurants are moderately to highly rated
- Customer trust depends on both ratings and number of votes
- Metro cities show high demand but intense competition
- Smaller cities show better service consistency
- Higher price does not guarantee higher satisfaction
- Cuisine choice strongly influences restaurant success



RECOMMENDATIONS

1. Food platforms should rank restaurants using both ratings and vote counts



2. Expansion strategies should consider high-performing smaller cities



3. Restaurant owners should focus on value for money instead of high pricing



4. Introducing niche cuisines can improve customer satisfaction and differentiation



SUMMARY



The restaurant dataset contains a large number of restaurant records across multiple cities in India. Exploratory Data Analysis helped in understanding customer behavior, rating trends, city-wise performance, and cuisine preferences. The analysis revealed that popularity, pricing balance, and location play a major role in restaurant success.

CONCLUSION

The restaurant dataset contains a large number of restaurant records across multiple cities in India. Exploratory Data Analysis helped in understanding customer behavior, rating trends, city-wise performance, and cuisine preferences. The analysis revealed that popularity, pricing balance, and location play a major role in restaurant success.



Thank YOU

FOR CHECKING THIS OUT

 **GITHUB:**

<https://github.com/mynkbhardwaj87007-spec/ZOMATO-ANALYSIS-INDIAN-RESTAURANT>

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